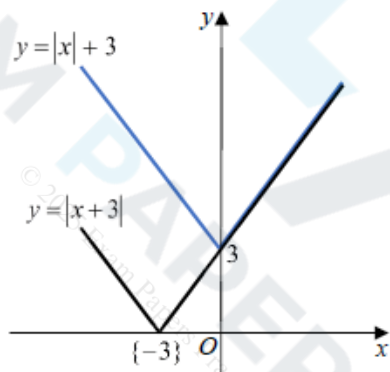


Mark Scheme

Q1.

Question	Scheme		Marks	AOs
	Statement: "If m and n are irrational numbers, where $m \neq n$, then mn is also irrational."			
(a)	E.g. $m = \sqrt{3}$, $n = \sqrt{12}$		M1	1.1b
	$\{mn = \} \quad (\sqrt{3})(\sqrt{12}) = 6$ $\Rightarrow$ statement untrue or 6 is not irrational or 6 is rational		A1	2.4
			(2)	
(b)(i), (ii) Way 1		V shaped graph {reasonably} symmetrical about the y-axis with vertical intercept (0, 3) or 3 stated or marked on the positive y-axis	B1	1.1b
		Superimposes the graph of $y = x + 3 $ on top of the graph of $y = x + 3$	M1	3.1a
	the graph of $y = x + 3$ is either the same or above the graph of $y = x + 3 $ {for corresponding values of x } or when $x \geq 0$, both graphs are equal (or the same) when $x < 0$, the graph of $y = x + 3$ is above the graph of $y = x + 3 $		A1	2.4
			(3)	
(b)(ii) Way 2	<u>Reason 1</u> When $x \geq 0$, $ x + 3 = x + 3 $	Any one of Reason 1 or Reason 2	M1	3.1a
	<u>Reason 2</u> When $x < 0$, $ x + 3 > x + 3 $	Both Reason 1 and Reason 2	A1	2.4
(5 marks)				

Notes for Question			
(a)			
M1:	States or uses any pair of <i>different</i> numbers that will disprove the statement. E.g. $\sqrt{3}$, $\sqrt{12}$; $\sqrt{2}$, $\sqrt{8}$; $\sqrt{5}$, $-\sqrt{5}$; $\frac{1}{\pi}$, 2π ; $3e$, $\frac{4}{5e}$;		
A1:	Uses correct reasoning to disprove the given statement, with a correct conclusion		
Note:	Writing $(3e)\left(\frac{4}{5e}\right) = \frac{12}{5} \Rightarrow$ untrue is sufficient for M1A1		
(b)(i)			
B1:	See scheme		
(b)(ii)			
M1:	For constructing a method of comparing $ x +3$ with $ x+3 $. See scheme.		
A1:	Explains fully why $ x +3 \geq x+3 $. See scheme.		
Note:	Do not allow either $x > 0$, $ x +3 \geq x+3 $ or $x \geq 0$, $ x +3 \geq x+3 $ as a valid reason		
Note	$x = 0$ (or where necessary $x = -3$) need to be considered in their solutions for A1		
Note:	Do not allow an incorrect statement such as $x \leq 0$, $ x +3 > x+3 $ for A1		
(b)(ii)			
Note:	Allow M1A1 for $x > 0$, $ x +3 = x+3 $ and for $x \leq 0$, $ x +3 \geq x+3 \geq$		
Note:	Allow M1 for any of x is positive, $x +3 = x+3$ x is negative, $x +3 > x+3$ $x > 0$, $x +3 = x+3$ $x \leq 0$, $x +3 \geq x+3$ $x > 0$, $x +3$ and $x+3$ are equal $x \geq 0$, $x +3$ and $x+3$ are equal - when $x \geq 0$, both graphs are equal - for positive values $x +3$ and $x+3$ are the same Condone for M1 $x \leq 0$, $x +3 > x+3$ $x < 0$, $x +3 \geq x+3$		
(b)(ii) Way 3	- For $x > 0$, $x +3 = x+3$ - For $-3 < x < 0$, as $x +3 > 3$ and $\{0 < \} x+3 < 3$, then $x +3 > x+3$ - For $x \leq -3$, as $x +3 = -x+3$ and $x+3 = -x-3$, then $x +3 > x+3$	M1	3.1a
		A1	2.4

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$2^x \times (2^2)^y = 2^{-\frac{3}{2}} \Rightarrow 2^{x+2y} = 2^{-\frac{3}{2}}$	M1	This mark is given for writing all terms in the same base and applying an index law
	$x + 2y = -\frac{3}{2}$	M1	This mark is given for writing an equation to link x and y
	$y = -\frac{1}{2}x - \frac{3}{4}$	A1	This mark is given for rearranging to find a correct expression of y as a function of x
(Total 3 marks)			

Q3.

Question	Scheme	Marks	AOs
(a)	$y \leq 7$	B1	2.5
		(1)	
(b)	$f(1.8) = 7 - 2 \times 1.8^2 = 0.52 \Rightarrow gf(1.8) = g(0.52) = \frac{3 \times 0.52}{5 \times 0.52 - 1} = \dots$	M1	1.1b
	$gf(1.8) = 0.975$ oe e.g. $\frac{39}{40}$	A1	1.1b
		(2)	
(c)	$y = \frac{3x}{5x-1} \Rightarrow 5xy - y = 3x \Rightarrow x(5y-3) = y$	M1	1.1b
	$(g^{-1}(x)) = \frac{x}{5x-3}$	A1	2.2a
		(2)	
(5 marks)			
Notes			
(a) B1: Correct range. Allow $f(x)$ or f for y. Allow e.g. $\{y \in \mathbb{R} : y \leq 7\}$, $-\infty < y \leq 7$, $(-\infty, 7]$ (b) M1: Full method to find $f(1.8)$ and substitutes the result into g to obtain a value. Also allow for an attempt to substitute $x = 1.8$ into an attempt at $gf(x)$. E.g. $gf(x) = \frac{3(7-2x^2)}{5(7-2x^2)-1} = \frac{3(7-2(1.8)^2)}{5(7-2 \times (1.8)^2)-1} = \dots$ A1: Correct value (c) M1: Correct attempt to cross multiply, followed by an attempt to factorise out x from an xy term and an x term. If they swap x and y at the start then it will be for an attempt to cross multiply followed by an attempt to factorise out y from an xy term and a y term. A1: Correct expression. Allow equivalent correct expressions e.g. $\frac{-x}{3-5x}$, $\frac{1}{5} + \frac{3}{25x-15}$ Ignore any domain if given.			

Q4.

Question	Scheme	Marks	AOs
	$15 - 2^{x+1} = 3 \times 2^x$	B1	1.1b
	$\Rightarrow 15 - 2 \times 2^x = 3 \times 2^x \Rightarrow 2^x = 3$ or e.g. $\Rightarrow \frac{15}{2^x} - 2 = 3 \Rightarrow 2^x = 3$	M1	1.1b
	$2^x = 3 \Rightarrow x = \dots$	dM1	1.1b
	$x = \log_2 3$	A1cso	1.1b
	(4)		
	Alternative		
	$y = 3 \times 2^x \Rightarrow 2^x = \frac{y}{3} \Rightarrow y = 15 - 2 \times \frac{y}{3}$	B1	1.1b
	$3y + 2y = 45 \Rightarrow y = 9 \Rightarrow 3 \times 2^x = 9 \Rightarrow 2^x = 3$	M1	1.1b
	$2^x = 3 \Rightarrow x = \dots$	dM1	1.1b
	$x = \log_2 3$	A1cso	1.1b
	(4 marks)		

Notes:

B1: Combines the equations to reach $15 - 2^{x+1} = 3 \times 2^x$ or equivalent e.g. $15 - 2^{x+1} - 3 \times 2^x = 0$

M1: Uses $2^{x+1} = 2 \times 2^x$ or e.g. $\frac{2^{x+1}}{2^x} = 2$ to obtain an equation in 2^x and attempts to make 2^x the subject.

See scheme but e.g. $y = 2^x \Rightarrow 3 \times 2^x = 15 - 2^{x+1} \Rightarrow 3y = 15 - 2y \Rightarrow y = \dots$ is also possible

dM1: Uses logs correctly and proceeds to a value for x from an equation of the form $2^x = k$ where $k > 1$

e.g. $2^x = k \Rightarrow x = \log_2 k$

or $2^x = k \Rightarrow \log 2^x = \log k \Rightarrow x \log 2 = \log k \Rightarrow x = \dots$

or $2^x = k \Rightarrow \ln 2^x = \ln k \Rightarrow x \ln 2 = \ln k \Rightarrow x = \dots$

Depends on the first method mark

This may be implied if they go straight to decimals e.g. $2^x = 3$ so $x = 1.584\dots$ but you may need to check

A1cso: $x = \log_2 3$ or $\frac{\log 3}{\log 2}$ or $\frac{\ln 3}{\ln 2}$

Ignore any attempts to find the y -coordinate

Alternative

B1: Correct equation in y

M1: Solves their equation in y and attempts to make 2^x the subject.

dM1: Uses logs correctly and proceeds to a value for x from an equation of the form $2^x = k$ where $k > 1$

e.g. $2^x = k \Rightarrow x = \log_2 k$

or $2^x = k \Rightarrow \log 2^x = \log k \Rightarrow x \log 2 = \log k \Rightarrow x = \dots$

or $2^x = k \Rightarrow \ln 2^x = \ln k \Rightarrow x \ln 2 = \ln k \Rightarrow x = \dots$

Depends on the first method mark

This may be implied if they go straight to decimals e.g. $2^x = 3$ so $x = 1.584\dots$ but you may need to check

A1cso: $x = \log_2 3$ or $\frac{\log 3}{\log 2}$ or $\frac{\ln 3}{\ln 2}$

Ignore any attempts to find the y -coordinate

Q5.

Question	Scheme	Marks	AOs
(a)	$\ln x \rightarrow \frac{1}{x}$	B1	1.1a
	Method to differentiate $\frac{4x^2 + x}{2\sqrt{x}}$ – see notes	M1	1.1b
	E.g. $2 \times \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$	A1	1.1b
	$\frac{dy}{dx} = 3\sqrt{x} + \frac{1}{4\sqrt{x}} - \frac{4}{x} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}}$ *	A1*	2.1
	(4)		
(b)	$12x^2 + x - 16\sqrt{x} = 0 \Rightarrow 12x^{\frac{3}{2}} + x^{\frac{1}{2}} - 16 = 0$	M1	1.1b
	E.g. $12x^{\frac{3}{2}} = 16 - \sqrt{x}$	dM1	1.1b
	$x^{\frac{3}{2}} = \frac{4}{3} - \frac{\sqrt{x}}{12} \Rightarrow x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12}\right)^{\frac{2}{3}}$ *	A1*	2.1
	(3)		
(c)	$x_2 = \sqrt[3]{\left(\frac{4}{3} - \frac{\sqrt{2}}{12}\right)^2}$	M1	1.1b
	$x_2 = \text{awrt } 1.13894$	A1	1.1b
	$x = 1.15650$	A1	2.2a
	(3)		
(10 marks)			

Notes:

(a)

B1: Differentiates $\ln x \rightarrow \frac{1}{x}$ seen or implied

M1: Correct method to differentiate $\frac{4x^2 + x}{2\sqrt{x}}$:

Look for $\frac{4x^2 + x}{2\sqrt{x}} \rightarrow \dots x^{\frac{3}{2}} + \dots x^{\frac{1}{2}}$ being then differentiated to $Px^{\frac{1}{2}} + \dots$ or $\dots + Qx^{-\frac{1}{2}}$

Alternatively uses the quotient rule on $\frac{4x^2 + x}{2\sqrt{x}}$.

Condone slips but if rule is not quoted expect $\left(\frac{dy}{dx}\right) = \frac{2\sqrt{x}(Ax+B) - (4x^2+x)Cx^{\frac{1}{2}}}{(2\sqrt{x})^2} (A, B, C > 0)$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.

Alternatively uses the product rule on $(4x^2 + x)(2\sqrt{x})^{-1}$

Condone slips but expect $\left(\frac{dy}{dx}\right) = Ax^{-\frac{1}{2}}(Bx+C) + D(4x^2+x)x^{-\frac{3}{2}} (A, B, C > 0)$

In general condone missing brackets for the M mark. If they quote $u = 4x^2 + x$ and $v = 2\sqrt{x}$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: Correct differentiation of $\frac{4x^2 + x}{2\sqrt{x}}$ although may not be simplified.

Examples: $\left(\frac{dy}{dx}\right) = \frac{2\sqrt{x}(8x+1) - (4x^2+x)x^{\frac{1}{2}}}{(2\sqrt{x})^2}, \frac{1}{2}x^{\frac{1}{2}}(8x+1) - \frac{1}{4}(4x^2+x)x^{\frac{3}{2}}, 2 \times \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$

A1*: Obtains $\frac{dy}{dx} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}}$ via $3\sqrt{x} + \frac{1}{4\sqrt{x}} - \frac{4}{x}$ or a correct application of the quotient or product rule

and with sufficient working shown to reach the printed answer.

There must be no errors e.g. missing brackets.

(b)

M1: Sets $12x^2 + x - 16\sqrt{x} = 0$ and divides by $\sqrt{x}$ or equivalent e.g. divides by x and multiplies by $\sqrt{x}$

dM1: Makes the term in $x^{\frac{3}{2}}$ the subject of the formula

A1*: A correct and rigorous argument leading to the given solution.

Alternative - working backwards:

$$x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12}\right)^{\frac{2}{3}} \Rightarrow x^{\frac{3}{2}} = \frac{4}{3} - \frac{\sqrt{x}}{12} \Rightarrow 12x^{\frac{3}{2}} = 16 - \sqrt{x} \Rightarrow 12x^2 = 16\sqrt{x} - x \Rightarrow 12x^2 - 16\sqrt{x} + x = 0$$

M1: For raising to power of $3/2$ both sides. **dM1:** Multiplies through by $\sqrt{x}$. **A1:** Achieves printed answer and makes a minimal comment e.g. tick, #, QED, true etc.

(c)

M1: Attempts to use the iterative formula with $x_1 = 2$. This is implied by sight of $x_2 = \left(\frac{4}{3} - \frac{\sqrt{2}}{12}\right)^{\frac{2}{3}}$ or awrt 1.14

A1: $x_2 =$ awrt 1.13894

A1: Deduces that $x = 1.15650$

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Q6.

Question	Scheme	Marks	AOs
(a)	Sets $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$	B1	1.1a
	Substitutes either $P=0$ or $P=\frac{11}{2}$ into $1 = A(11-2P) + BP \Rightarrow A \text{ or } B$	M1	1.1b
	$\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$	A1	1.1b
		(3)	
(b)	Separates the variables $\int \frac{22}{P(11-2P)} dP = \int 1 dt$	B1	3.1a
	Uses (a) and attempts to integrate $\int \frac{2}{P} + \frac{4}{(11-2P)} dP = t + c$	M1	1.1b
	$2 \ln P - 2 \ln(11-2P) = t + c$	A1	1.1b
	Substitutes $t=0, P=1 \Rightarrow t=0, P=1 \Rightarrow c = (-2 \ln 9)$	M1	3.1a
	Substitutes $P=2 \Rightarrow t = 2 \ln 2 + 2 \ln 9 - 2 \ln 7$	M1	3.1a
	Time = 1.89 years	A1	3.2a
		(6)	

(c)	Uses ln laws $2 \ln P - 2 \ln(11-2P) = t - 2 \ln 9$ $\Rightarrow \ln\left(\frac{9P}{11-2P}\right) = \frac{1}{2}t$	M1	2.1
	Makes 'P' the subject $\Rightarrow \left(\frac{9P}{11-2P}\right) = e^{\frac{1}{2}t}$ $\Rightarrow 9P = (11-2P)e^{\frac{1}{2}t}$ $\Rightarrow P = f\left(e^{\frac{1}{2}t}\right) \text{ or } \Rightarrow P = f\left(e^{\frac{1}{2}t}\right)$	M1	2.1
	$\Rightarrow P = \frac{11}{2 + 9e^{\frac{1}{2}t}} \Rightarrow A=11, B=2, C=9$	A1	1.1b
		(3)	
(12 marks)			

**Notes:**

(a)

B1: Sets $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$

M1: Substitutes $P=0$ or $P=\frac{11}{2}$ into $1 = A(11-2P) + BP \Rightarrow A$ or B

Alternatively compares terms to set up and solve two simultaneous equations in A and B

A1: $\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$ or equivalent $\frac{1}{P(11-2P)} = \frac{1}{11P} + \frac{2}{11(11-2P)}$

Note: The correct answer with no working scores all three marks.

(b)

B1: Separates the variables to reach $\int \frac{22}{P(11-2P)} dP = \int 1 dt$ or equivalent

M1: Uses part (a) and $\int \frac{A}{P} + \frac{B}{(11-2P)} dP = A \ln P \pm C \ln(11-2P)$

A1: Integrates both sides to form a correct equation including a 'c' Eg
 $2 \ln P - 2 \ln(11-2P) = t + c$

M1: Substitutes $t=0$ and $P=1$ to find c

M1: Substitutes $P=2$ to find t . This is dependent upon having scored both previous M's

A1: Time = 1.89 years

(c)

M1: Uses correct log laws to move from $2 \ln P - 2 \ln(11-2P) = t + c$ to $\ln\left(\frac{P}{11-2P}\right) = \frac{1}{2}t + d$ for their numerical 'c'

M1: Uses a correct method to get P in terms of $e^{\frac{1}{2}t}$

This can be achieved from $\ln\left(\frac{P}{11-2P}\right) = \frac{1}{2}t + d \Rightarrow \frac{P}{11-2P} = e^{\frac{1}{2}t+d}$ followed by cross multiplication and collection of terms in P (See scheme)

Alternatively uses a correct method to get P in terms of $e^{-\frac{1}{2}t}$. For example

$\frac{P}{11-2P} = e^{\frac{1}{2}t+d} \Rightarrow \frac{11-2P}{P} = e^{-\left(\frac{1}{2}t+d\right)} \Rightarrow \frac{11}{P} - 2 = e^{-\left(\frac{1}{2}t+d\right)} \Rightarrow \frac{11}{P} = 2 + e^{-\left(\frac{1}{2}t+d\right)}$ followed by division

A1: Achieves the correct answer in the form required. $P = \frac{11}{2 + 9e^{-\frac{1}{2}t}} \Rightarrow A=11, B=2, C=9$ oe

Q7.

Question	Scheme	Marks	AOs
(a)	$\sqrt{\frac{1+4x}{1-x}} = (1+4x)^{0.5} \times (1-x)^{-0.5}$	B1	3.1a
	$(1+4x)^{0.5} = 1 + 0.5 \times (4x) + \frac{0.5 \times -0.5}{2} \times (4x)^2$	M1	1.1b
	$(1-x)^{-0.5} = 1 + (-0.5)(-x) + \frac{(-0.5) \times (-1.5)}{2} (-x)^2$	M1	1.1b
	$(1+4x)^{0.5} = 1 + 2x - 2x^2$ and $(1-x)^{-0.5} = 1 + 0.5x + 0.375x^2$ oe	A1	1.1b
	$(1+4x)^{0.5} \times (1-x)^{-0.5} = (1+2x-2x^2 \dots) \times \left(1 + \frac{1}{2}x + \frac{3}{8}x^2 \dots\right)$ $= 1 + \frac{1}{2}x + \frac{3}{8}x^2 + 2x + x^2 - 2x^2 + \dots$ $= A + Bx + Cx^2$	dM1	2.1
	$= 1 + \frac{5}{2}x - \frac{5}{8}x^2 \dots$ *	A1*	1.1b
		(6)	
(b)	Expression is valid $ x < \frac{1}{4}$ Should not use $x = \frac{1}{2}$ as $\frac{1}{2} > \frac{1}{4}$	B1	2.3
		(1)	
(c)	Substitutes $x = \frac{1}{11}$ into $\sqrt{\frac{1+4x}{1-x}} \approx 1 + \frac{5}{2}x - \frac{5}{8}x^2$	M1	1.1b
	$\sqrt{\frac{5}{2}} = \frac{1183}{968}$	A1	1.1b
	(so $\sqrt{6}$ is) $\frac{1183}{484}$ or $\frac{2904}{1183}$	A1	2.1
		(3)	
(10 marks)			

(a)

B1: Scored for key step in setting up the process so that it can be attempted using binomial expansions

This could be achieved by $\sqrt{\frac{1+4x}{1-x}} = (1+4x)^{0.5} \times (1-x)^{-0.5}$ See end for other alternatives

It may be implied by later work.

M1: Award for an attempt at the binomial expansion $(1+4x)^{0.5} = 1 + 0.5 \times (4x) + \frac{(0.5) \times (-0.5)}{2} \times (4x)^2$

There must be three (or more terms). Allow a missing bracket on the $(4x)^2$ and a sign slip so the correct application may be implied by $1 + 2x \pm 0.5x^2$

M1: Award for an attempt at the binomial expansion $(1-x)^{-0.5} = 1 + (-0.5)(-x) + \frac{(-0.5) \times (-1.5)}{2} (-x)^2$

There must be three (or more terms). Allow a missing bracket on the $(-x)^2$ and a sign slips so the method may be awarded on $1 \pm 0.5x \pm 0.375x^2$

A1: Both correct and simplified. This may be awarded for a correct final answer if a candidate does all their simplification at the end

dM1: In the main scheme it is for multiplying their two expansions to reach a quadratic. It is for the key step in adding 'six' terms to produce the quadratic expression. Higher power terms may be seen. Condone slips on

the multiplication on one term only. It is dependent upon having scored the first B and one of the other two M's

In the alternative it is for multiplying $\left(1 + \frac{5}{2}x - \frac{5}{8}x^2\right)(1-x)^{0.5}$ and comparing it to $(1+4x)^{0.5}$

It is for the key step in adding 'six' terms to produce the quadratic expression.

A1*: Completes proof with no errors or omissions. In the alternative there must be some reference to the fact that both sides are equal.

(b)

B1: States that the expansion may not / is not valid when $|x| > \frac{1}{4}$

This may be implied by a statement such as $\frac{1}{2} > \frac{1}{4}$ or stating that the expansion is only valid when $|x| < \frac{1}{4}$

Condone, for this mark a candidate who substitutes $x = \frac{1}{2}$ into the $4x$ and states it is not valid as $2 > 1$ or

Don't award for candidates who state that $\frac{1}{2}$ is too big without any reference to the validity of the expansion.

As a rule you should see some reference to $\frac{1}{4}$ or $4x$

(c)(i)

M1: Substitutes $x = \frac{1}{11}$ into BOTH sides $\sqrt{\frac{1+4x}{1-x}} \approx 1 + \frac{5}{2}x - \frac{5}{8}x^2$ and attempts to find at least one side.

As the left hand side is $\frac{\sqrt{6}}{2}$ they may multiply by 2 first which is acceptable

A1: Finds both sides leading to a correct equation/statement $\sqrt{\frac{15}{10}} = \frac{1183}{968}$ or $\sqrt{6} = 2 \times \frac{1183}{968}$

A1: $\sqrt{6} = \frac{1183}{484}$ or $\sqrt{6} = \frac{2904}{1183}$ $\sqrt{6} = 2 \times \frac{1183}{968} = \frac{1183}{484}$ would imply all 3 marks

Watch for other equally valid alternatives for 11(a) including

B1: $(1+4x)^{0.5} \approx \left(1 + \frac{5}{2}x - \frac{5}{8}x^2\right)(1-x)^{0.5}$ then the M's are for $(1+4x)^{0.5}$ and $(1-x)^{0.5}$

M1: $(1-x)^{0.5} = 1 + (0.5)(-x) + \frac{(0.5) \times (-0.5)}{2}(-x)^2$

Or

B1: $\sqrt{\frac{1+4x}{1-x}} = \sqrt{1 + \frac{5x}{1-x}} = \left(1 + 5x(1-x)^{-1}\right)^{\frac{1}{2}}$ then the first M1 for one application of binomial and the second would be for both $(1-x)^{-1}$ and $(1-x)^{-2}$

Or

B1: $\sqrt{\frac{1+4x}{1-x}} \times \frac{\sqrt{1-x}}{\sqrt{1-x}} = \sqrt{(1+3x-4x^2)} \times (1-x)^{-1} = \left(1 + (3x-4x^2)\right)^{\frac{1}{2}} \times (1-x)^{-1}$